

Project Standard

Process Numbering System

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1. General

The purpose of the instruction is to describe the numbering system for process areas, equipment, motors, instruments, pipelines and hand valves.

The item numbers as described within this document will be shown in flowsheets, equipment specifications, drawings, cost calculations, purchase specifications etc., and must therefore be well known to all personnel related to a project, from a pre- study to a completed delivery project.

All items within the project will be designated by a numerical code consisting of several parts.

Some projects have an overall project code. This code is effective for all equipment and software tagging and does not need to be showed in documents and drawings.

2. Process Area codes

Master list	Department	Department	Department	Department	Department	Department	Department	Department	Department	Department
Block 0 Common systems	00 x	01 x	02x	03x	04x	05x	06x	07x	08x	09x
	0	0	0	0	0	0	0	0	0	0
	1	1	1	1	1	1	1	1	1	1
	2	2	2	2	2	2	2	2	2	2
	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4	4	4
	5	5	5	5	5	5	5	5	5	5
	6	6	6	6	6	6	6	6	6	6
	7	7	7	7	7	7	7	7	7	7
	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	
Block 1 Paper machine no.1	10 x	11 x	12x	13x	14x	15x	16x	17x	18x	19x
	0	0, Paper machine common	0, Bale handling system	0, Pulp mixing & Approach system	0, Power distribution	0, Winder	0, Mist removal system	0, Tissue machine	0, Fresh water system	0, DIP Reject handling
	1	1	1, Pulp dissolving system	1	1	1	1	1 TAD 1	1	1, DIP Pulping & Coarse screening
	2	2	2, Softwood line	2, White water system	2, Boiler plant	2	2, Advantage aircap & air system	2 TAD 2	2, Compressed air system	2, DIP Flotation
	3	3	3	3	3	3	3	3	3	3, DIP Cleaning & Fine screening
	4	4	4, Hardwood line	4, Shower water system	4, Steam & Condensate system	4, Roll handling & Wrapping	4, Advantage wet dust system PM & Advantage run	4 Web Inspection	4, Internal effluent treatment system	4, DIP Thickening & First loop water system
	5	5	5	5	5	5	5	5, Control system	5	5, DIP Dispersing
	6	6	6, Internal broke system	6, Vacuum system	6, Co-generation plant	6	6, Dust control for rewinder	6, Control power distribution	6, Gas system	6, DIP Bleaching
	7	7	7	7	7, Overhead crane	7	7	7, Emergency Stop system	7	7, DIP Washing & Clarification
	8	8	8, Converting broke system	8	8, External effluent treatment	8	8, Machine hall ventilation	8 DMM, Machine Monitoring System	8, TM Chemical system	8, DIP Reductive bleaching
9	9	9	9	9	9	9	9, Spark detection system	9, DIP Chemical system	9	

	20 x	21 x	22x	23x	24x	25x	26x	27x	28x	29x
Block 2 Paper machine no.2	0	0, Paper machine common	0, Bale handling system	0, Pulp mixing & Approach system	0, Power distribution	0, Winder	0, Mist removal system	0, Tissue machine	0, Fresh water system	0, DIP Reject handling
	1	1	1, Pulp dissolving system	1	1	1	1	1 TAD 1	1	1, DIP Pulping & Coarse screening
	2	2	2, Softwood line	2, White water system	2, Boiler plant	2	2, Advantage aircap & air system	2 TAD 2	2, Compressed air system	2, DIP Flotation
	3	3	3	3	3	3	3	3	3	3, DIP Cleaning & Fine screening
	4	4	4, Hardwood line	4, Shower water system	4, Steam & Condensate system	4, Roll handling & Wrapping	4, Advantage wet dust system PM & Advantage run	4 Web Inspection	4, Internal effluent treatment system	4, DIP Thickening & First loop water system
	5	5	5	5	5	5	5	5, Control system	5	5, DIP Dispersing
	6	6	6, Internal broke system	6, Vacuum system	6, Co-generation plant	6	6, Dust control for rewinder	6, Control power distribution	6, Gas system	6, DIP Bleaching
	7	7	7	7	7, Overhead crane	7	7	7, Emergency Stop system	7	7, DIP Washing & Clarification
	8	8	8, Converting broke system	8	8, External effluent treatment	8	8, Machine hall ventilation	8 DMM, Machine Monitoring System	8, TM Chemical system	8, DIP Reductive bleaching
	9	9	9	9	9	9	9	9, Spark detection system	9, DIP Chemical system	9
	40 x	41 x	42x	43x	44x	45x	46x	47x	48x	49x
Block 3, 4, 5 etc.	0	0	0	0	0	0	0	0	0	0
	1	1	1	1	1	1	1	1	1	1
	2	2	2	2	2	2	2	2	2	2
	3	3	3	3	3	3	3	3	3	3
	4	4	4	4	4	4	4	4	4	4
	5	5	5	5	5	5	5	5	5	5
	6	6	6	6	6	6	6	6	6	6
	7	7	7	7	7	7	7	7	7	7
	8	8	8	8	8	8	8	8	8	8
	9	9	9	9	9	9	9	9	9	9

First digit – Block. (for example, 1 = Paper machine no.1)

Digit two and three- Department / Area code. (for example, 22 = Stock preparation softwood line)

The Paper Island is divided in following sub systems, process areas:

10-General common

60-Mist removal system

62-Advantage Aircap and air system

64 Advantage Wet Dust system PM & Advantage run

66-Dust control for rewinder

68-Machine hall ventilation

20-Bale Handling system

70-Paper Machine

21-Pulp dissolving system

71-TAD 1

22-Softwood line

72-TAD 2

24-Hardwood line

74-Web Inspection

26-Internal broke system

75-Control System

28-Converting broke system

76-Control power distribution

77-Emergency Stop system

78-DMM, Machine Monitoring System

79-Spark detection system

30-Pulp mixing & Approach system

80-Fresh water system

32-White water system

82-Compressed air system

34-Shower water system

84-Internal Effluent treatment system

36-Vacuum system

86-Gas system

88-TM Chemical system

89-DIP Chemical system

40-Power distribution

90-DIP Reject handling

42-Boiler plant

91-DIP Pulping & Coarse screening

44-Steam & Condensate system

92-DIP Flotation

46-Co-generation plant

93-DIP Cleaning & Fine screening

47-Over head crane

94-DIP Thickening & 1-loop water system

48-External effluent treatment

95-DIP Dispersing

96-DIP Bleaching

97-DIP Washing & Clarification

98-DIP Reductive bleaching

50-Winder

54-Roll Handling & Wrapping

3 Equipment numbering system

All equipment within the project is designated by a numerical code, the position number consisting of four parts:

xyz-aaa

x One digit Mill area code (PM1 = 1)

yy Two digit process area code (see section 7)

z One letter to designate the process equipment function (see section 9)

aaa An individual running number from 000 to 999 for each process equipment function within each area.

3.1 Numbering of machines

Example:

Softwood pulper 122E-001

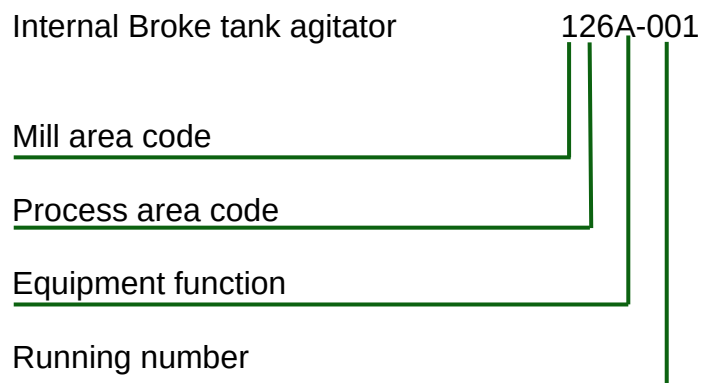
Mill area code

Process area code

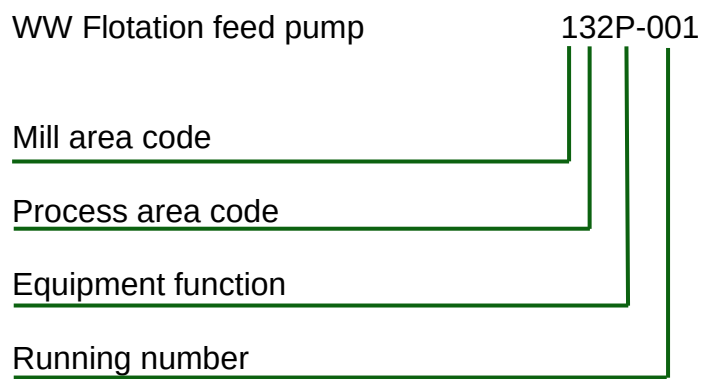
Equipment function

Running number

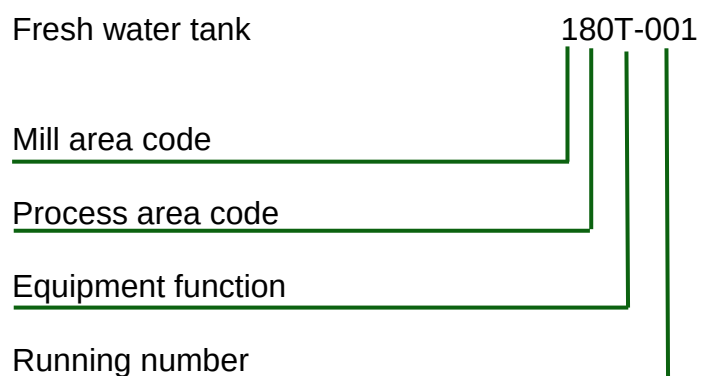
3.2 Numbering of agitators



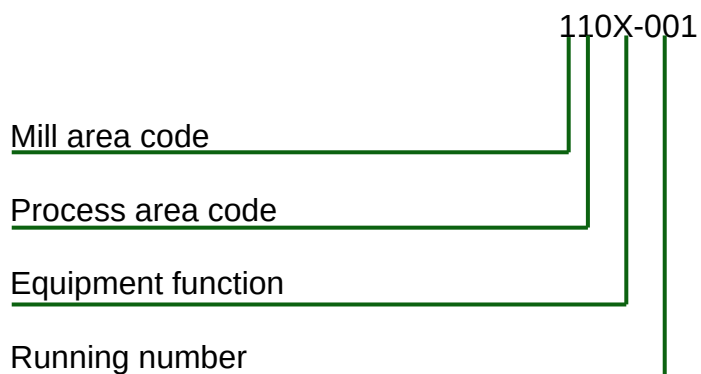
3.3 Numbering of pumps



3.4 Numbering of tanks



3.5 Numbering of other



3.6 Numbering of motors

All motors in the project are designated by a numerical code consisting of two parts, the equipment position number and the motor number. Note that a machine can have several motors, in general the main motor is always denominated as M1.

Example equipment with one motor:

Hardwood pulper pump	124P-001-M1
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Example equipment with several motors e.g. Softwood refiner:

Softwood refiner, main motor	122E-002-M1
Softwood refiner, oil pump motor	122E-002-M2
Softwood refiner, loading motor	122E-002-M3

3.7 Numbering of manual valves

All manual valves in the project are designated by a numerical code consisting of four parts.

xyyV-aaa

x One-digit Mill area code (PM1 = 1)

yy Two-digit process area code (see section 7)

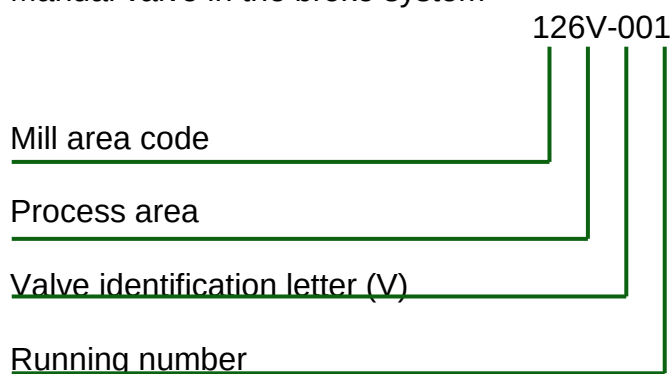
V- Letter V identifies the number as a valve identification.

Aaa - An individual running number from 001 to 999 for each process sub system.

Valve type name is according to SSG 7531E

Example:

Manual valve in the broke system



3.8 Numbering of pipelines

All pipelines in the project are designated by a numerical code consisting of four parts

xyyL-aaa

x One-digit Mill area code (PM1 = 1)

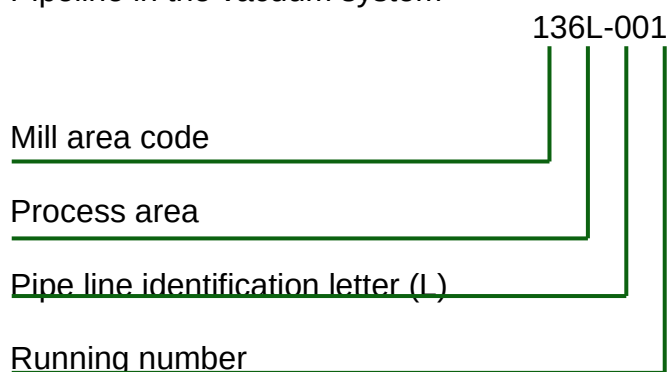
yy Two-digit process area code

L- Letter L identifies the number as a pipe line identification.

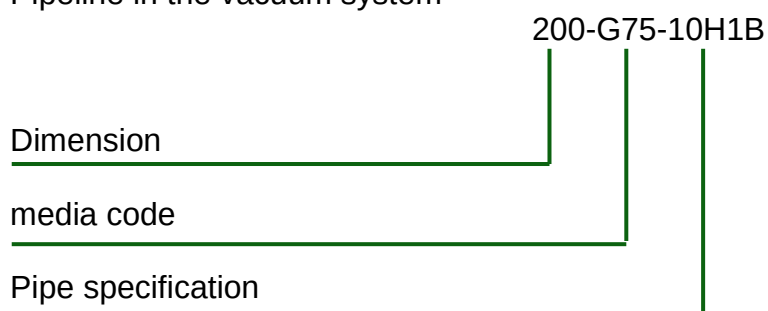
aaa- An individual running number from 001 to 999 for each process sub system.

Example:

Pipeline in the vacuum system



Pipeline in the vacuum system



3.9 Tubing numbering system

Hydraulic and pneumatic tubes are numbered from actuator to source.

The tubes will have the same name as the unit it goes from (circuit, valve panel etc.)

3.9.1 Actuators to HP, VB

Tubes from actuators to HP, VB is numbered as:

BCC-DEE

B	machine section
CC	circuit number
D	type of tube
	H = hydraulic
	Dr = hydraulic drain
	P = pneumatic
EE	serial number for this circuit

example 190-H1 (section 1, circuit 90, hydraulic tube no. 1)
122-P2 (section 1, circuit 22, pneumatic tube no. 2)

3.9.2 HP, VB to source

Tubes to valve panels and boxes from hydraulic units or pneumatic supply lines is numbered as:

YYBCC-DEE

YY	device code for boxes.
B	machine section
CC	circuit number
D	type of tube
	P = hydraulic pressure line
	T = hydraulic return line
	Dr = hydraulic drain line
	P = pneumatic
EE	serial number for this circuit

example VB228-P1 (to valve panel VB228, pressure tube 1(from hydraulic unit))
VB121-P1 (to valve box VB121, pneumatic tube 1 (from pneumatic supply line))

3.9.3 Numbering system for tubes for lube. system.

CCC-DE(-FF)

CCC roll number or flow panel number
D type of tube

L = lube oil pressure line to bearing
LR = lube oil return line
LP = lube oil supply line

E = serial number
FF = machine side (if applicable)

TS=tending side
DS=drive side

example	1#01-L1-TS	(from lube panel to roll 1#01 tending side)
	2#03-LR1-DS	(return from roll 2#03 drive side)
	LP2-LP	(supply to lube panel 2)
	LR2	(return header 2)

Generally: name (roll, panels etc.) to left and running number to right of tube code

4 Equipment codes for control units and locations.

XYZZ-BCC

X	Tissue Machine Number, set by customer where area is the entire PM
YY	area number set by customer where area is the entire PM
ZZ	Device code (acc. to list below)
BCC	Machine section/Running number

ex 170CB-110 PM,control box, wet end, box no 10

ZZ DEVICE/UNIT/LOCATION CODES

CB	control box
CD	control desk
CC	control cabinet
JB	junction box
EB	equipment box (for electric equipment)
EC	equipment cabinet (for electric equipment)
VB	valve box
HP	hydraulic valve panel
HU	hydraulic power unit or heating unit
LU	central lubrication unit
LP	lube flow meter panel
TS	tending side
DS	drive side

B the machine section e.g. (TM)

- 0 = auxiliary systems
- 1 = wet end (Headbox+Former+TAD+Press)
- 2 = dry end (Yankee+Calender+Reel)
- 3 = core and roll handling
- 4 = steam & condensate
- 5 = wet end hood & burner
- 6 = dry end hood & burner

CC a serial number for the machine section
Valmet standard division of numbers is as follows. (Even numbers on TS, Odd on DS).

Auxiliary devices (CD, HU, HP, LU etc)	x00-x09
Control boxes (CB)	x10-x19
Valve boxes (VB)	x20-x39
Junction boxes (JB)	x40-x59
Control Centers (CC)	x60-x79
Equipment Cabinets (EC)	x80-x89

5 Numbering of control loops and instruments

The position code (tag name) format is based on the following principles:

Control loop	XYYZZZ-ABCC
Position code	XYYZZZN-ABCC
Signal name	XYYZZZN-ABCCD

where:

X	=	Tissue Machine X (set by the project)
YY	=	Area code (set by the project)
ZZZ	=	device code (see below)
N	=	running number for device.
ABCC	=	circuit number

where:

A	=	process area code or 1000-series number (optional)
B	=	machine section codes (see list below)
CC	=	running circuit number (see list below)
D	=	running letter when several control signals on same device

Explanation

X = "Tissue Machine" is set by the project

YY = "Area code" is set by the project and should have max 2 digits

e.g. 342 = stock prep. area
344 = paper machine controls

ZZZ = "device code". Letter codes for components according to ISO

e.g. PT = pressure transmitter
GSV = solenoid valve

N = "running number for device". Even if there is only one PT in a loop that one will be PT1. The next ones will be PT2, PT3 etc.

A = "process area code" is set by the project and should have max 1 digit. Normally this is the first digit in a 1000-series number given by the customer for the use of the supplier (eg loop numbers to be used by supplier 3000-3499). This number may be left out.

B = "section code". These numbers are set by Valmet based on the scope of supply.

e.g.

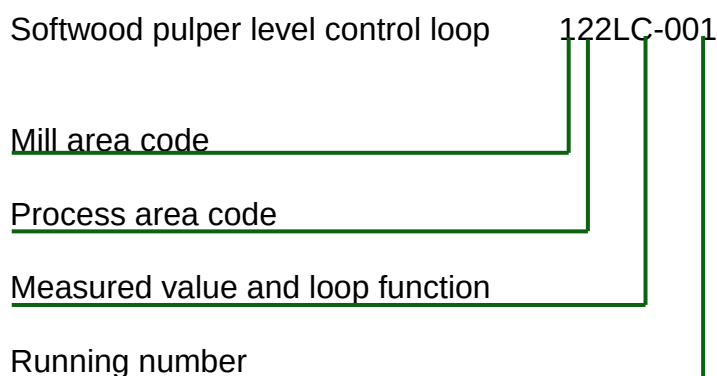
- 0 = auxiliary systems (hydraulics, lube, etc.)
- 1 = wet end (headbox + forming section + TAD/Press section)
- 2 = dry end (Yankee + sheet transfer + calender + reel)
- 3 = core and roll handling
- 4 = steam & condensate
- 5 = wet end hood & burner
- 6 = dry end hood & burner

CC = "running number" within section

D = running letter when there are several control signals on the same device. E.g. a solenoid valve with two spools or a switch with several contacts.

170GSV1-3115A	first valve, first solenoid
170GSV1-3115B	first valve, second solenoid
170GSV2-3115	second valve, one solenoid

Example:



The level control transmitter and the level control valve will in the instrument lists consequently are named as:

122LT1-001	(Level transmitter)
122LV1-001	(Level control valve)

In case a control loop consists of several transmitters or valves the equipment is named as:

126LT1-001	(Level transmitter)
126LV1-001	(Level control valve no 1)
126LV2-001	(Level control valve no 2)

5.1 Identification letters for Instrumentation (ISO 14617 Based)

(Device code ZZZ)

L e t t e r	Group 1		Group 2	
	Measured or initial variable		Function	Instrument
	Meaning as first letter	Meaning as modifier	Meaning as succeeding letter	
	(1)	(2)	(3)	(4)
A	Analysis	-	Alarm	Annunciator
B	-	-	-	-
C	Conductivity	-	Controlling	Controller
D	Density, Directional valves	Difference	Quick exhaust	-
E	All electrical variables	-	-	Primary element
F	Flow	Ratio	-	-
G	Gauging, position or length	-	-	-
H	Hand or direct interlock operated	-	High	-
I	Monitoring	-	Indicating	Indicator, On-off indication
J	Power	-	-	-
K	Time or time programme or sequence	-	-	-
L	Level	-	Low	-
M	Moisture or humidity	-	-	Signal converter, proportional valves
N	-	-	-	-
O	-	-	-	-
P	Pressure or vacuum	-	-	Test point
Q	Quality and analysis, for example pH, Consistency	Integrate or totalize	Integrating or summing	Integrating or summing device
R	Radioactive	-	Recording	Recorder or printer
S	Speed	-	Switching	Switch
T	Temperature	-	Transmitting	Transmitter
U	Multivariable	-	-	Multifunction unit
V	Viscosity	-	-	Valve, damper, etc.
W	Weight or force	-	-	Well, process connection
X	Unclassified variables	-	Unclassified variables	Computing relay or relay
Y	User's choice	-	-	-
Z	Quantity	-	Emergency or safety acting	Emergency or safety device

Explanation:

Group 1

The measured or initiating variables denoted by the first letter are given in Column 1.

The meaning of the first letter can be modified by the addition of a letter from Column 2.

Group 2

The letters identifying functions are given in Column 3.

Column 4 gives the functions when the succeeding letters identify specific instruments themselves, e.g. LV = level control valve, TT = temperature transmitter, PS = pressure switch, FE = flow element (mag tube or orifice), PE = process connection for pressure measurement in piping, etc.

Identification letters for electrification (IEC 1346-1)

(Device code ZZZ)

	Item	Examples
A	Assemblies, Subassemblies	-
B	Transducers, from non-electrical to electrical quantity or vice versa.	-
C	Capacitors	-
D	Binary elements, delay devices, storage devices	-
E	Miscellaneous	-
F	Protective devices	Fuses
G	Generators, Power supplies	-
H	Signalling devices	Optical indicator, acoustic indicator.
J	-	-
K	Relays contractors	-
L	Inductors, reactors	-
M	Motors	-
N	Analogue elements	-
P	Measuring equipment, testing equipment	-
Q	Switching devices for power circuits,	Circuit breaker, Disconnecter
R	Resistors	-
S	Switching devices for control circuits selectors	Control switch, push-button, limit switch, selector switch, dial contact, connecting stage
T	Transformers	Voltage transformer, current transformer
U	Modulators, changers	-
V	Tubes, semiconductors	-
W	Transmission paths	Conductor, cable, busbar etc.
X	Terminals, plugs, sockets	-
Y	Electrically operated mechanical devices.	Brake, Clutch
Z	Terminations, hybrid transformers, filters, equalizers, limiters	-

5.2 Typical device codes used by Valmet for electrical/Instrument devices.

DEVICE	CODE	DEVICE	CODE
Control switches/Push Buttons	HS	Auxiliary relays	KA
Illuminated pushbuttons	HSI	Emergency Relay	KZ
Proximity/Limit switches/ Photocells	GS	Power units	T
Pilot lights	HI	Miniature Circuit breakers (MCB)	F
		Molded Case Circuit Breaker (MCCB)	Q
Flow Indicator (Local)	FI	Fuses	FU
Position/Length Indicator (Local)	GI		
Pressure Gauge	PI	Pressure/Vacuum Switch	PS
Temperature Indicator	TI	Flow Switch	FS
Weight/Force	WI	Level Switch	LS
Speed/Frequency	SI	Temperature Switch	TS
Flow Sensor	FE		
Position/length Sensor	GE	Electrical brake	Y
Pressure Sensor	PE	Electric motor	M
Temperature Sensor	TE	Over voltage protector	V
Load cells	WE	Heater	H
Speed/frequency Sensor	SE	Contactor	K
Consistency Sensor	QE	Wall socket	X
Analyser Sensor	AE		
		Stabilisator	Z
Flow Transmitter	FT		
Position/Length Transmitter	GT	LC filter	Z
Pressure Transmitter	PT	Rack	A
Temperature Transmitter	TT	Harting-connector	EX
Load Transmitter	WT	Lamp	GX
Analyzer Transmitter	AT		
Optical trigger Transmitter	RT		
Vibration Transmitter	XT		
Consistency Transmitter	QT		

5.3 Typical device codes used by Valmet for fluid devices.

DEVICE	CODE
On-off valves (Automated)	HV
Pressure control valve	PCV
Pressure proportional valves	PCM
Flow control valves	FCV
Check valves/Manual Valves (Mill)	V
Check valves/Manual Valves (TM)	HV
Level Control Valve	LCV
Temperature Control Valve	TCV
Consistency Control Valve	QCV
Solenoid valves	GSV
I/P-converters	PM
Pressure Reducing Valve (Local)	PRV
Pressure Safety Valve	PSV
Sequence Valve	KV

5.4 AC - Motor control I/O - tags for process and machine automation system parts

	Normal drive	Reverse drive	Reverse drive 2 speed	Variable drive	Variable reverse drive	Electric heater
	ND	RD	RD/2S	VD	VD/RD	Heating
Run command	MC	MCF	MCF	VMC	VMCF	MC
Run feedback	RF	RF	RF	VRF	VRF	RF
Motor fault	MF	MF	MF	VMF	VMF	MF
Motor ready	RY	RY	RY	VRV	VRV	RY
Run reverse		MCR	MCR		VMCR	
High speed			MCH			
Low speed			MCL			
Speed setpoint				VSF	VSF	
Current measurement	CI	CI	CI	CI	CI	CI